Southern Nuclear

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Docket Nos.: 50-366 NL-24-0050

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001

Edwin I. Hatch Nuclear Plant – Unit 2
Licensee Event Report 2024-001-00
Unit 2 High Pressure Coolant Injection (HPCI) System Inoperable

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(v)(D), Southern Nuclear Operating Company hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact the Hatch Licensing Manager, Jimmy Collins, at 912.453.2342.

Respectfully submitted,

M.S. Busch
Vice President – Hatch

MSB/JMH

Enclosure: LER 2024-001-00

Cc: NRC Regional Administrator, Region II
NRR Project Manager – Plant Hatch
NRC Senior Resident Inspector – Plant Hatch
RTYPE: CHA02.004

Edwin I. Hatch Nuclear Plant Unit 2
Licensee Event Report 2024-001-00
Unit 2 High Pressure Coolant Injection (HPCI) System Inoperable

Enclosure

LER 2024-001-00

NRC FORM 366 (10-01-2023)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 03/31/2024

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Edwin I. Hatch Nuclear Plant Unit 2	[X] 050 / [ ] 052	00366	1 OF 2

4. Title

Unit 2 High Pressure Coolant Injection (HPCI) System Inoperable

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	[ ] 050	Docket Number
12	18	2023	2024	- 001 -	00	02	14	2024	Facility Name	[ ] 052	Docket Number

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	[ ] 20.2203(a)(2)(vi)	**10 CFR Part 50**	[ ] 50.73(a)(2)(ii)(A)	[ ] 50.73(a)(2)(viii)(A)	[ ] 73.1200(a)
[ ] 20.2201(b)	[ ] 20.2203(a)(3)(i)	[ ] 50.36(c)(1)(i)(A)	[ ] 50.73(a)(2)(ii)(B)	[ ] 50.73(a)(2)(viii)(B)	[ ] 73.1200(b)
[ ] 20.2201(d)	[ ] 20.2203(a)(3)(ii)	[ ] 50.36(c)(1)(ii)(A)	[ ] 50.73(a)(2)(iii)	[ ] 50.73(a)(2)(ix)(A)	[ ] 73.1200(c)
[ ] 20.2203(a)(1)	[ ] 20.2203(a)(4)	[ ] 50.36(c)(2)	[ ] 50.73(a)(2)(iv)(A)	[ ] 50.73(a)(2)(x)	[ ] 73.1200(d)
[ ] 20.2203(a)(2)(i)	**10 CFR Part 21**	[ ] 50.46(a)(3)(ii)	[ ] 50.73(a)(2)(v)(A)	**10 CFR Part 73**	[ ] 73.1200(e)
[ ] 20.2203(a)(2)(ii)	[ ] 21.2(c)	[ ] 50.69(g)	[ ] 50.73(a)(2)(v)(B)	[ ] 73.77(a)(1)	[ ] 73.1200(f)
[ ] 20.2203(a)(2)(iii)		[ ] 50.73(a)(2)(i)(A)	[ ] 50.73(a)(2)(v)(C)	[ ] 73.77(a)(2)(i)	[ ] 73.1200(g)
[ ] 20.2203(a)(2)(iv)		[ ] 50.73(a)(2)(i)(B)	[X] 50.73(a)(2)(v)(D)	[ ] 73.77(a)(2)(ii)	[ ] 73.1200(h)
[ ] 20.2203(a)(2)(v)		[ ] 50.73(a)(2)(i)(C)	[ ] 50.73(a)(2)(vii)		

[ ] **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Jimmy Collins - Licensing Manager	9124532342

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	BN	TIS	G082	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
[X] No	[ ] Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

At 0223 EST on 12/18/2023, while Unit 2 was at 100% power in MODE 1, the High Pressure Coolant Injection (HPCI) outboard steam isolation valve received a spurious close signal resulting in the HPCI system being isolated and unavailable to automatically start. Consequently, HPCI was declared inoperable.

The cause of HPCI isolating was electronic noise stemming from the age-related failure of a Reactor Core Isolation Cooling (RCIC) Analog Transmitter Trip System (ATTS) card. The manner in which the RCIC ATTS card failed did not adversely affect the operability of RCIC. However, due to the proximity of the RCIC ATTS card and the HPCI ATTS card (both located on the same instrument rack), the electronic noise caused the HPCI ATTS card to introduce a spurious signal to isolate the HPCI system. The RCIC System and the low pressure Emergency Core Cooling Systems (ECCS) were operable and available during this event.

Subsequently, the failed RCIC ATTS card was replaced and operability of the HPCI System was restored at 0512 EST on 12/18/2023 by resetting the isolation signal and returning the system to standby line up.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Edwin I. Hatch Nuclear Plant Unit 2	■ 050 ☐ 052	00366	2024	001	00

NARRATIVE

EVENT DESCRIPTION
At 0223 EST on 12/18/2023, while Unit 2 was at 100% power in MODE 1, the High Pressure Coolant Injection (HPCI) (EIIS: BJ) outboard steam isolation valve received a spurious close signal resulting in the HPCI system being isolated and unavailable to automatically start. Consequently, HPCI was declared inoperable. The spurious HPCI isolation was due to electronic noise stemming from a failed Reactor Core Isolation Cooling (RCIC) (EIIS: BN) Analog Transmitter Trip System (ATTS) card (EIIS: TIS). The RCIC ATTS card was replaced and HPCI was restored to operable status at 0512 EST on 12/18/2023. The RCIC System and the low pressure Emergency Core Cooling Systems (ECCS) were operable and available during this event.

FAILED COMPONENTS INFORMATION:
Master Parts List Number: 2E51N666B
Manufacturer: General Electric
Model Number: GE P/N 184C5988G201
Type: Analog Transmitter Trip System (ATTS)

EVENT CAUSE ANALYSIS
The cause of HPCI isolating was electronic noise stemming from the age-related failure of a RCIC ATTS card. The manner in which the RCIC ATTS card failed did not adversely affect the operability of RCIC. However, due to the proximity of the RCIC ATTS card and the HPCI ATTS card (both located on the same instrument rack), the electronic noise caused the HPCI ATTS card to introduce a spurious signal to isolate the HPCI system.

SAFETY ASSESSMENT
Even though HPCI lost its ability to automatically start, the RCIC System and the low pressure ECCS were operable and available during this event. However, because HPCI does not have a redundant system the isolation of the HPCI System and loss of its automatic functions represents a condition that could have prevented the fulfillment of the HPCI safety function to mitigate the consequences of an accident and is reportable per 50.73(a)(2)(v)(D). There were no actual safety consequences as a result of this event. The event was within the analysis of the UFSAR Chapter 15.

CORRECTIVE ACTIONS
The failed RCIC ATTS card was successfully replaced and operability for the HPCI System was restored at 0512 EST on 12/18/2023 by resetting the isolation signal and returning the system to standby line up. An extent of condition is being performed to identify other ATTS cards located in the same panels that could potentially be susceptilble to failure due to electronic noise. Additionally, preventative maintenance strategies are being created to replace ATTS cards prior to age-related failure.

PREVIOUS SIMILAR EVENTS
On March 28, 2023, with Unit 1 at 100% rated thermal power, an operability review concluded that the High Pressure Coolant Injection (HPCI) pump would not have automatically started during a 22 minute period on March 18, 2023 when an invalid high suppression pool water level signal was received and before the signal was reset. Causal analysis determined that the failure of an unrelated instrument card, located on the same instrument rack as the instrument card monitoring Unit 1 suppression pool water level, caused a momentary voltage perturbation on the rack, likely leading to an invalid suppression pool water level high signal from the suppression pool water level instrument card. This high-level signal caused the Unit 1 HPCI system trip solenoid to actuate. There was no actual high level in the Unit 1 suppression pool. Corrective actions included resetting the invalid suppression pool level high signal and replacing the failed instrument card.